SOP-LV-05 — Monitoring & Alerting Setup

Kode	SOP-LV-05
Pemilik	Levi (Implementor / DevOps / SRE)
Revisi	1.0 · **Berlaku** 2026-05-29 · **Reviu** per kuartal
Referensi	Google SRE (SLO/SLI), ITIL 4 Monitoring & Event Mgmt, tools/monitoring-config.md
COBIT	DSS01 (Managed Operations)

1. Tujuan

Pasang **"CCTV + alarm"** buat tiap service — observability (log/metric/trace) + alert yang **matter** — supaya incident **kedeteksi sebelum user komplain**, bukan sesudah. Alert yang noise = sama buruknya dengan gak ada alert.

2. Ruang Lingkup

- **Berlaku:** tiap service baru sebelum go-live, dan tiap service production existing (kontrol LV-C4: gak ada go-live tanpa monitoring+alert).
- **Tidak berlaku:** prototype lokal/staging non-production.

3. Definisi & Istilah

- **SLI** — Service Level Indicator: angka ukur kesehatan (% request sukses, latency p99, error rate).
- **SLO** — Service Level Objective: target SLI (mis. "99.5% request sukses").
- **Error budget** — sisa jatah error sebelum nabrak SLO; abis → freeze deploy.
- **Alert threshold** — ambang yang kalau dilewatin → alarm nyala.
- **Alert fatigue** — kebanyakan alarm palsu → orang ngabaikan alarm beneran.

4. Referensi

Google SRE (4 golden signals: latency, traffic, errors, saturation), 12-Factor (log as event stream), mandate no silent auto (alert eksplisit), kontrol LV-C4.

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Tentukan SLI/SLO	Levi	Levi	Kakashi, @lelouch (target bisnis)	Tata
Pasang log/metric/trace	Levi	Levi	@saitama (hook di kode)	—
Set alert threshold	Levi	Levi	—	tim
Review alert noise	Levi	Levi	—	@nami

6. Prosedur

6.1 Definisi sinyal

- 6.1.1 Tentukan **SLI** per service (4 golden signals: latency, traffic, errors, saturation).
- 6.1.2 Tetapkan **SLO** (target) — konsultasi @kakashi + @lelouch (angka bisnis, mis. "user gak nunggu > 3 detik").
- 6.1.3 Tentukan **error budget** dari SLO.

6.2 Instrumentasi

- 6.2.1 Pasang **structured log** (JSON + request id + user id) — koord @saitama (hook di BE).
- 6.2.2 Pasang **metric collection** (mis. Prometheus) + **dashboard** (mis. Grafana).
- 6.2.3 Pasang **health check** endpoint (`/health` return state komponen).
- 6.2.4 (Kalau perlu) **tracing** (OpenTelemetry) buat request lintas-servis.

6.3 Alerting

- 6.3.1 Set **alert threshold** untuk metric kritis (error rate, latency p99, saturation) berdasar SLO.
- 6.3.2 Alert **actionable** — tiap alert ada runbook "kalau nyala, lakuin apa". Alert tanpa aksi = noise → hapus.
- 6.3.3 Routing: alert → channel + oncall. Failure = noisy, success = silent (no spam).
- 6.3.4 **Decision point:** alert noise ratio tinggi? → tune threshold / hapus alert non-actionable (anti alert-fatigue).
- 6.3.5 **Exit criteria:** SLI/SLO tertulis + dashboard live + alert kritis aktif & actionable + health check return 200 + (kalau go-live) tercentang di pre-flight kontrol LV-C4.

7. Pengecualian

- **Service sementara / spike experiment:** monitoring minimal (health + error rate) cukup, gak perlu full SLO — tapi tetap ada alert dasar.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
SLI/SLO definition	monitoring-config / `log.md`	Living
Dashboard + alert config	repo / platform monitoring	Living
Bukti aktif sebelum go-live	pre-flight checklist	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
MTTD	menit incident mulai → alert nyala	minimal (alert duluan dari komplain)
Alert noise ratio	# alert false/non-actionable ÷ total alert	rendah
Monitoring coverage	# service go-live dengan monitoring+alert ÷ total	100% (kontrol LV-C4)

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama